

- 1 Discuss feature extraction power of your favorite CNN pretrained by ImageNet dataset before and after transfer learning by MNIST digit dataset by plotting high dimensional feature vectors on 2D plane using three dimension reduction techniques.

We have used three dimension reduction techniques PCA, t-SNE, and UMAP.

PCA (Principal Component Analysis) is a technique that reduces data dimensions by capturing the directions with the most variance.

Before transfer learning: The PCA plot shows that the digit classes are not clearly separated, many of them overlap. This is because the model has not yet learned what makes each digit unique.

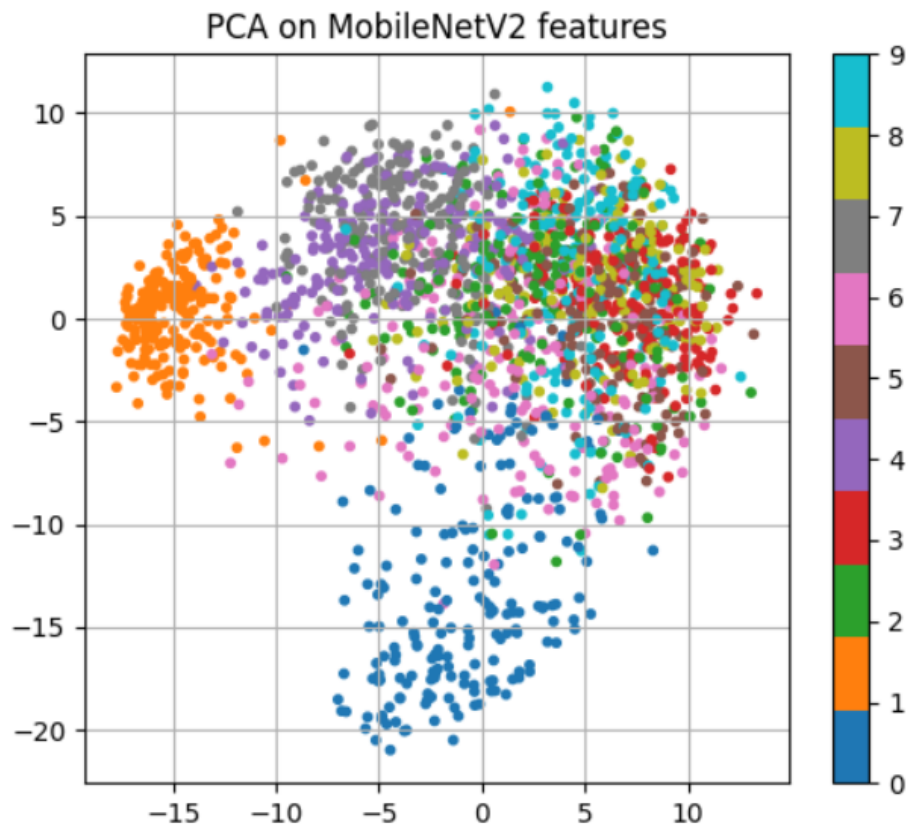


Figure 1: PCA before transfer learning

After transfer learning: The digits form clearer and more separate clusters in the PCA plot. This means the model is now better at finding the important features that help tell digits apart.

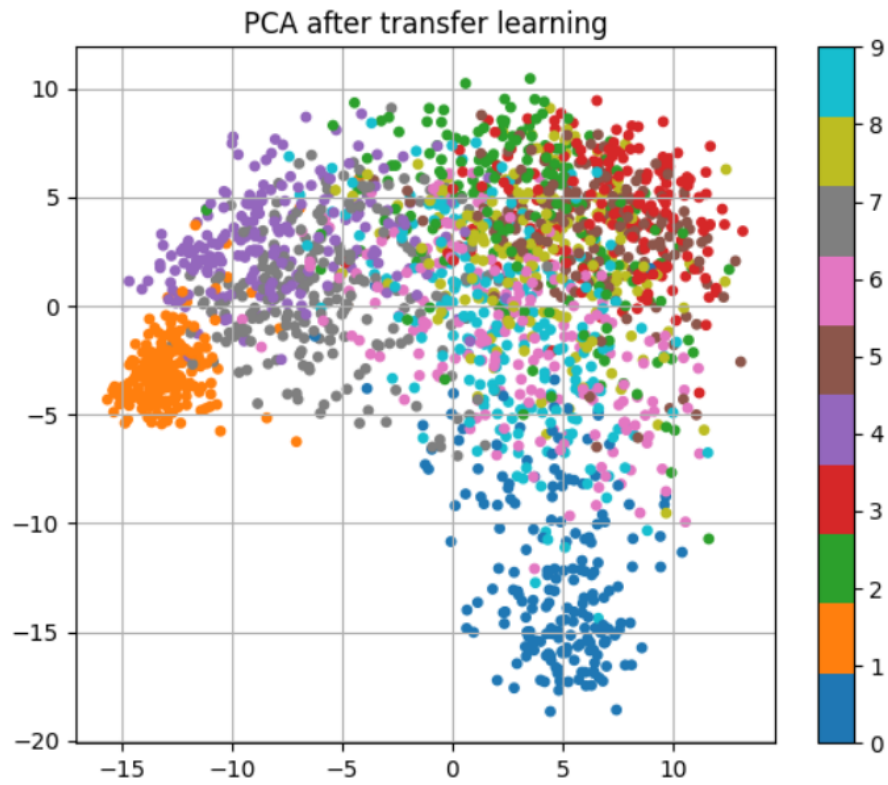


Figure 2: PCA after transfer learning

t-SNE (t-distributed Stochastic Neighbor Embedding) is a technique that focuses on keeping similar data points close together in a lower-dimensional space.

Before transfer learning: The t-SNE plot shows unclear and scattered clusters, with many digits blending into each other. This means the features extracted by the model are not very useful for telling different digits apart.

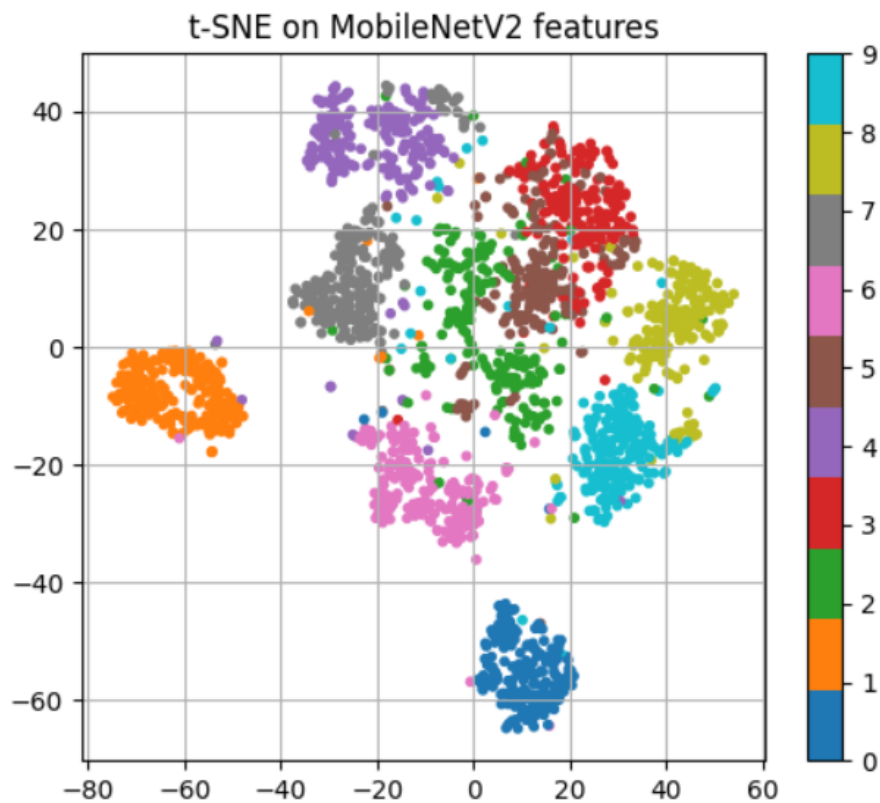


Figure 3: t-SNE before transfer learning

After transfer learning:, the t-SNE plot shows tight and well-separated clusters, with each digit forming its own group.

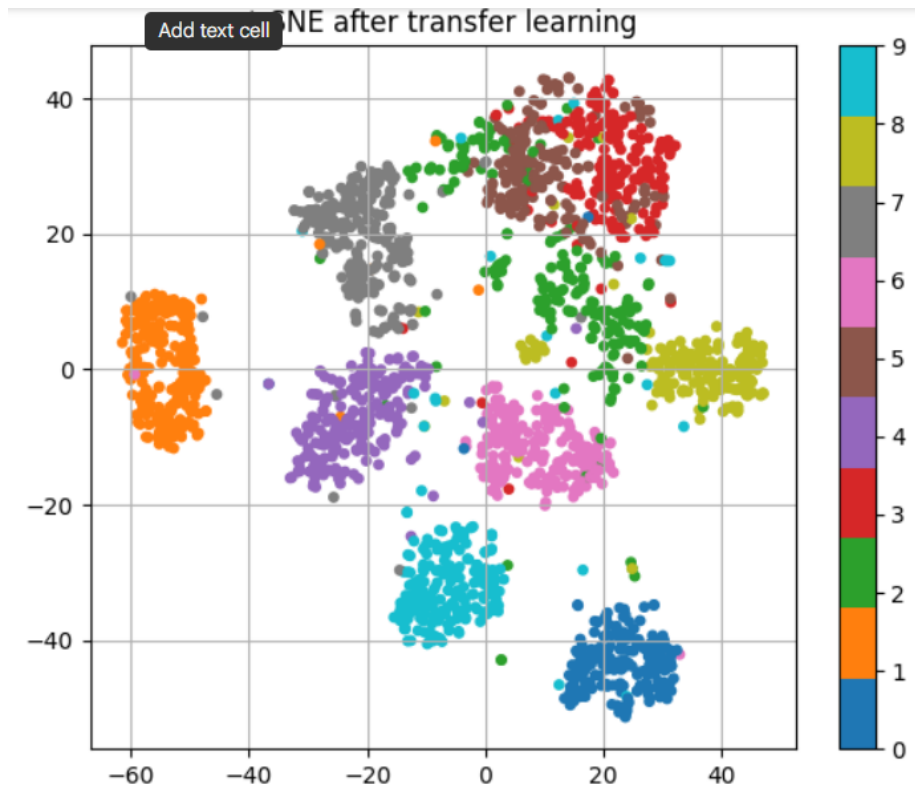


Figure 4: t-SNE after transfer learning

UMAP (Uniform Manifold Approximation and Projection) is a method that preserves both local and global structure when reducing dimensions.

Before transfer learning: Before transfer learning, the UMAP plot shows loose and overlapping clusters, where many digit classes are not clearly separated.

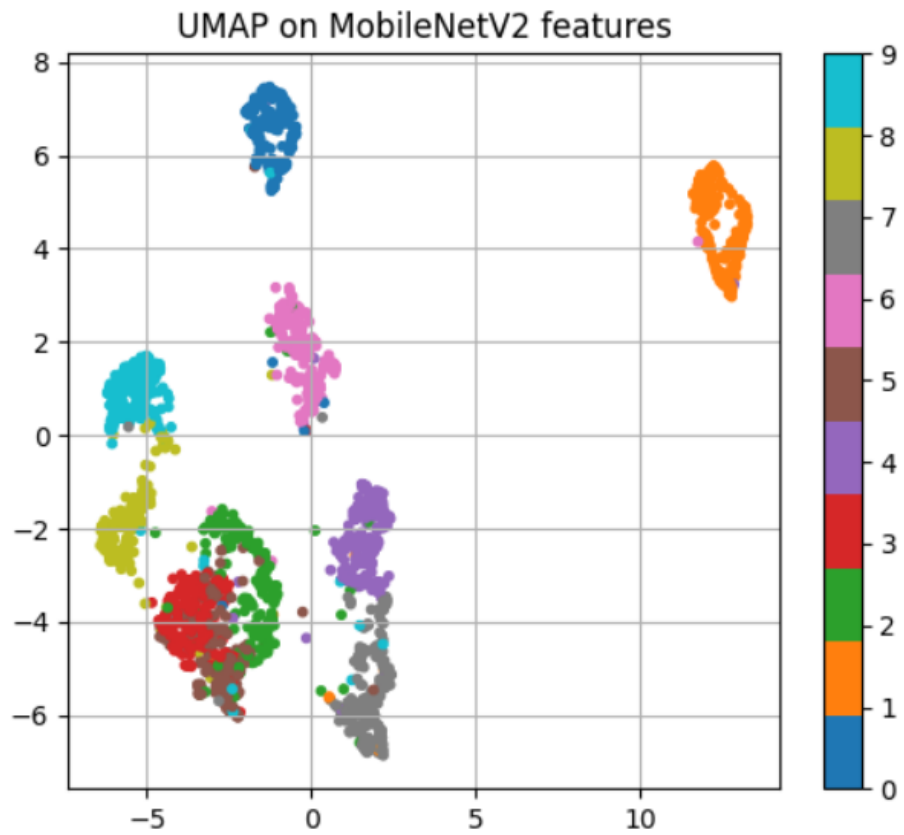


Figure 5: UMAP before transfer learning

After transfer learning:the digit classes are grouped into tight, well-separated clusters, showing that the model has developed a better understanding of the differences between digits.

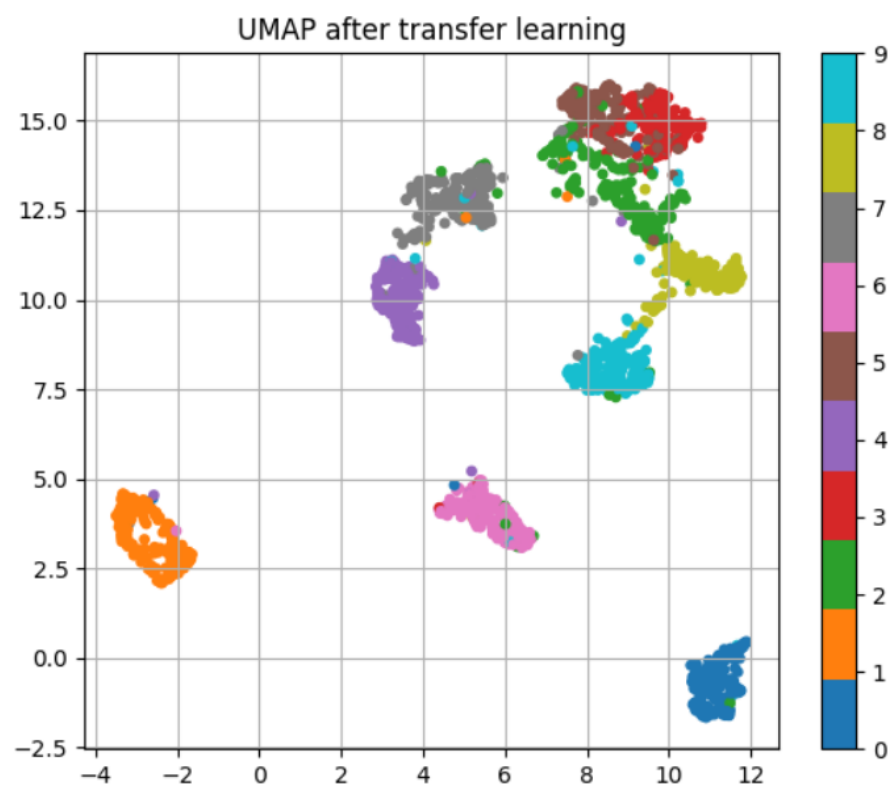


Figure 6: UMAP after transfer learning